HOMEWORK 5

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4.1. Find the inverse of the matrices

$$\begin{pmatrix} 1 & 2 & 1 \\ 3 & 7 & 3 \\ 2 & 3 & 4 \end{pmatrix}, \qquad \begin{pmatrix} 1 & -1 & 2 \\ 1 & 1 & -2 \\ 1 & 1 & 4 \end{pmatrix}.$$

Show all steps.

Proof. $R_2 \to R_2 - 3R_1; R_3 \to R_3 - 2R_1; R_3 \to R_3 + R_2; R_3 \to \frac{1}{2}R_3; R_1 \to R_1 - R_3; R_1 \to R_1 - 2R_2;$

$$A^{-1} = \begin{pmatrix} \frac{19}{2} & -\frac{5}{2} & -\frac{1}{2} \\ -3 & 1 & 0 \\ -\frac{5}{2} & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

$R_2 \to R_2 - R_1; R_3 \to R_3 - R_1; R_3 \to R_3 - R_2; R_2 \to \frac{1}{2}R_2; R_3 \to \frac{1}{6}R_3; R_1 \to R_1 + R_2; R_2 \to R_2 + 2R_3;$

$$A^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ -\frac{1}{2} & \frac{1}{6} & \frac{1}{3} \\ 0 & -\frac{1}{6} & \frac{1}{6} \end{pmatrix}$$

□

5.1. True or false:

a) Every vector space that is generated by a finite set has a basis;
b) Every vector space has a (finite) basis;
c) A vector space cannot have more than one basis;
d) If a vector space has a finite basis, then the number of vectors in every basis is the same.
e) The dimension of $\mathbb{P}_n$ is n;
f) The dimension on $M_{m\times n}$ is $m + n$;
g) If vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ generate (span) the vector space V, then every vector in V can be written as a linear combination of vector $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in only one way;
h) Every subspace of a finite-dimensional space is finite-dimensional;
i) If V is a vector space having dimension n, then V has exactly one subspace of dimension 0 and exactly one subspace of dimension n.

Proof. TFFTFFFTT □

5.2. Prove that if V is a vector space having dimension n, then a system of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in V is linearly independent if and only if it spans V.

Proof. Suppose $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are linearly independent, but do not span V. Thus, there exists $\mathbf{w} \notin \text{span}\{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$. Thus, $\mathbf{w}$ is linearly independent. Now, we may write $a_1\mathbf{v}_1 + \ldots + a_2\mathbf{v}_n + b\mathbf{w}$. Thus, by definition, $b = 0$. If it werent, we'd get $\mathbf{w} = -\frac{1}{b}(a_1\mathbf{v}_1 + \ldots + a_n\mathbf{v}_n)$, which contradicts the fact that $\mathbf{w}$ is independent. Thus, we have $n+1$ vectors now. But there can be at most n independent vectors in V. Thus, $v_1, \ldots, v_n$ span V.

Suppose $\mathbf{v}_1, \ldots, \mathbf{v}_n$ span V, but are not linearly independent. Thus, there is some vector v_i that we may remove from the span, giving us $n-1$ vectors. But in order to span V, the spanning set must have at least n vectors. Thus, $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are linearly independent. □

5.3. Prove that a linearly independent system of vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ in a vector space V is a basis if and only if $n = \dim V$.

Proof. We are given $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ linearly independent vectors in V. Now say that this system spans V. Thus, this is a basis for V. By definition, $\dim V = n$, because there are exactly n vectors in our basis that generate V. Conversely, assume $\dim V = n$. We must show that our linearly independent vectors span V. Hence, suppose there exists $\mathbf{w} \in V$ such that $\mathbf{w} \notin \text{span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$. Suppose $a_1\mathbf{v}_1 + \mathbf{v}_2 + \ldots + a_n\mathbf{v}_n + b\mathbf{w} = 0$. Assume $b \neq 0$. We get $\mathbf{w} = -\frac{1}{b}\sum_{i=1}^{n} a_i\mathbf{v}_i$. But we said $\mathbf{w}$ isn't in the span. Thus, $b = 0$. This shows $\mathbf{w}$ is linearly independent. Thus, we get $a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_n\mathbf{v}_n + b\mathbf{w} = 0$. Now, we have $n+1$ linearly independent vectors. But $\dim V = n$, and there can be at most n linearly independent vectors in V. This is a contradiction. Therefore, our original system $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ spans V, making it a basis for V. □

5.4. (An old problem revisited: now this problem should be easy) Is it possible that vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent, but the vectors $\mathbf{w}_1 = \mathbf{v}_1 + \mathbf{v}_2$, $\mathbf{w}_2 = \mathbf{v}_2 + \mathbf{v}_3$ and $\mathbf{w}_3 = \mathbf{v}_3 + \mathbf{v}_1$ are linearly *independent*? Hint: What dimension can the subspace $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ have?

Proof. No. Since $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are linearly dependent, they are linear combinations of the others, so $\dim(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3) \leq 2$. Since $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ are linear combinations of $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$, they lie in $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$. Since there are three vectors in an, at most, 2 dimensional space, one of them must be linearly dependent. Therefore, $\mathbf{w}_1, \mathbf{w}_2, \mathbf{w}_3$ cannot be linearly independent. □

5.5. Let vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be a basis in V. Show that $\mathbf{u} + \mathbf{v} + \mathbf{w}$, $\mathbf{v} + \mathbf{w}$, $\mathbf{w}$ is also a basis in V.

Proof. Suppose $a(u+v+w) + b(v+w) + cw = 0$. Then $au + (a+b)v + (a+b+c)w = 0$, so by linear independence of u, v, w, $a = a+b = a+b+c = 0$. Hence $u+v+w$, $v+w$, w are linearly independent. They also span V, since $u = (u+v+w) - (v+w)$, $v = (v+w) - w$, $w = w$. Thus they form a basis. □

5.6. Consider in the space $\mathbb{R}^5$ vectors $\mathbf{v}_1 = (2, -1, 1, 5, -3)^T$, $\mathbf{v}_2 = (3, -2, 0, 0, 0)^T$, $\mathbf{v}_3 = (1, 1, 50, -921, 0)^T$.

a) Prove that these vectors are linearly independent.
b) Complete this system of vectors to a basis.

If you do part b) first you can do everything without any computations.

Proof. By the Rank Theorem, we can construct the matrix and row reduce: $\begin{pmatrix} 1 & 1 & 50 & -921 & 0 \\ 3 & -2 & 0 & 0 & 0 \\ 2 & -1 & 1 & 5 & 3 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 50 & -921 & 0 \\ 0 & -5 & -150 & 2763 & 0 \\ 0 & 0 & -45 & 946 & 15 \end{pmatrix}$. Thus, by rank theorem, we see that the three vectors are linearly independent. Completing the basis, we get $\begin{pmatrix} 1 & 1 & 50 & -921 & 0 \\ 0 & -5 & -150 & 2763 & 0 \\ 0 & 0 & -45 & 946 & 15 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$. Thus, the basis is $\{(2,-1,1,5,-3)^T, (3,-2,0,0,0)^T, (1,1,50,-921,0)^T, (0,0,0,1,0)^T, (0,0,0,0,1)^T\}$ □

6.1.

a) Any system of linear equations has at least one solution;
b) Any system of linear equations has at most one solution;
c) Any homogeneous system of linear equations has at least one solution;
d) Any system of n linear equations in n unknowns has at least one solution;
e) Any system of n linear equations in n unknowns has at most one solution;
f) If the homogeneous system corresponding to a given system of a linear equations has a solution, then the given system has a solution;
g) If the coefficient matrix of a homogeneous system of n linear equations in n unknowns is invertible, then the system has no non-zero solution;
h) The solution set of any system of m equations in n unknowns is a subspace in $\mathbb{R}^n$;
i) The solution set of any homogeneous system of m equations in n unknowns is a subspace in $\mathbb{R}^n$.

Proof. FFTFFFTFT □

6.2. Find a 2×3 system (2 equations with 3 unknowns) such that its general solution has a form

$$\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + s \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \qquad s \in \mathbb{R}.$$

Proof. $\begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -2 \end{bmatrix} \mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. □

7.1.

a) The rank of a matrix is equal to the number of its non-zero columns;
b) The $m \times n$ zero matrix is the only $m \times n$ matrix having rank 0;
c) Elementary row operations preserve rank;
d) Elementary column operations do not necessarily preserve rank;
e) The rank of a matrix is equal to the maximum number of linearly independent columns in the matrix;
f) The rank of a matrix is equal to the maximum number of linearly independent rows in the matrix;
g) The rank of an $n \times n$ matrix is at most n;
h) An $n \times n$ matrix having rank n is invertible.

Proof. FTTFTTTT □

7.2. A 54×37 matrix has rank 31. What are dimensions of all 4 fundamental subspaces?

Proof. Column Space: 31
Null Space: 6
Row Space: 31
Left Null Space: 23 □

7.3. Compute rank and find bases of all four fundamental subspaces for the matrices

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{pmatrix}, \qquad \begin{pmatrix} 1 & 2 & 3 & 1 & 1 \\ 1 & 4 & 0 & 1 & 2 \\ 0 & 2 & -3 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

Proof. For first matrix:
Column Space: 2
Null space: 1
Row Space: 2
Left null space: 1
Ran A: $(1,0,1)^T,\ (1,1,1)^T$ (pivot columns of A).
Ran A^T: $(1,0,-1),\ (0,1,1)$.
Ker A: $(1,-1,1)^T$.
Ker A^T: $(1,0,-1)^T$.
Second matrix:
Ran B: $(1,1,0,1)^T,\ (2,4,2,0)^T,\ (3,0,-3,0)^T$ (pivot columns of B).
Ran B^T: $(1,0,0,0,0),\ (0,2,3,1,1),\ (0,0,-6,-1,0)$.
Ker B: $(0,-3,-2,12,0)^T,\ (0,-1,0,0,2)^T$.
Ker B^T: $(1,-1,1,0)^T$ (since $\text{row}_3 = \text{row}_2 - \text{row}_1$).

Col Space: 3
Null space: 2
Row Space: 3
Left Null Space : 1 □

7.4. Prove that if $A : X \to Y$ and V is a subspace of X then $\dim AV \le \operatorname{rank} A$. ($AV$ here means the subspace V transformed by the transformation A, i.e. any vector in AV can be represented as $A\mathbf{v}$, $\mathbf{v} \in V$). Deduce from here that $\operatorname{rank}(AB) \le \operatorname{rank} A$.

Remark: Here one can use the fact that if $V \subset W$ then $\dim V \le \dim W$. Do you understand why is it true?

Proof. AV is a subspace: $\mathbf{0} = A\mathbf{0} \in AV$, and $\alpha A\mathbf{v}_1 + \beta A\mathbf{v}_2 = A(\alpha\mathbf{v}_1 + \beta\mathbf{v}_2) \in AV$ since $\alpha\mathbf{v}_1 + \beta\mathbf{v}_2 \in V$. If $\mathbf{w} \in AV$, then $\mathbf{w} = A\mathbf{v}$ with $\mathbf{v} \in V \subset X$, so $\mathbf{w} \in \operatorname{Ran} A$. Thus $AV \subset \operatorname{Ran} A$, and $\dim AV \le \dim \operatorname{Ran} A = \operatorname{rank} A$.

For the second claim, $\operatorname{Ran}(AB) = A(\operatorname{Ran} B)$, so taking $V = \operatorname{Ran} B$ gives $\operatorname{rank}(AB) = \dim A(\operatorname{Ran} B) \le \operatorname{rank} A$. □

7.5. Prove that if $A : X \to Y$ and V is a subspace of X then $\dim AV \leq \dim V$. Deduce from here that $\operatorname{rank}(AB) \leq \operatorname{rank} B$.

Proof. Since V is a v.s., $\dim(V) = n \implies n$ vectors needed in basis. Let $v_1, \ldots, v_n$ be a basis for V. Thus, any $v \in V$ may be represented as $v = \sum_{i=1}^{n} \alpha_i v_i$. For any $w \in AV$, there exists some v such that $w = Av$. Thus, $w = \sum_{i=1}^{n} \alpha_i A(v_i)$. Thus, it takes n vectors to span AV, which shows $\dim(AV) \leq \dim(V)$. Doing the same process, we arrive at the same result for $\operatorname{Rank}(AB) \leq \operatorname{Rank}(B)$ when we let $V = \operatorname{Ran}(B)$. □

7.6. Prove that if the product AB of two $n \times n$ matrices is invertible, then both A and B are invertible. Even if you know about determinants, do not use them, we did not cover them yet. Hint: use previous 2 problems.

Proof. Since AB is an invertible $n \times n$ matrix, $\operatorname{rank}(AB) = n$. By 7.4 and 7.5,

$$n = \operatorname{rank}(AB) \leq \operatorname{rank} A \leq n, \qquad n = \operatorname{rank}(AB) \leq \operatorname{rank} B \leq n,$$

so $\operatorname{rank} A = \operatorname{rank} B = n$. An $n \times n$ matrix of rank n is invertible, so A and B are invertible. □

7.7. Prove that if $A\mathbf{x} = \mathbf{0}$ has unique solution, then the equation $A^T\mathbf{x} = \mathbf{b}$ has a solution for every right side $\mathbf{b}$.

Hint: count pivots

Proof. Let A be $m \times n$. Since $A\mathbf{x} = \mathbf{0}$ has a unique solution, there are no free variables, so the echelon form of A has a pivot in every column. Hence $\operatorname{rank} A = n$, and so $\operatorname{rank} A^T = \operatorname{rank} A = n$. But A^T is $n \times m$, so its n pivots lie in n different rows. Therefore $A^T\mathbf{x} = \mathbf{b}$ is consistent for every $\mathbf{b}$. □